

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

Playful AI: Intelligent Board Game Opponents and Advisors

Team Name:

GAME ANALYZER

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

Develop AI-powered opponents and advisors for board games that can adapt to player skill levels, provide strategic advice, and offer a playful, engaging experience.

Key Points:

1. Problem Statement:

- New players often struggle to learn complex strategies without guidance.
- Players want a more interactive and fun experience with AI opponents that mimic human-like behavior.
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2. Proposed Solution:

- An AI system that can act as both an opponent and an advisor in board games.
- The AI adapts to the player's skill level, providing a challenging yet fair experience.
- The AI offers real-time strategic advice to help players improve their gameplay.
- The AI has a playful personality, making the game more engaging and fun.

3. Target Users:

- Board game enthusiasts looking for intelligent opponents.
- New players seeking guidance to learn game strategies
- Casual players who want a fun, interactive experience with AI.

4. Expected Outcome:

- A functional AI system that can play board games, adapt to player skill levels, and provide strategic advice.
- A playful and engaging AI personality that enhances the gaming experience.

Phase-2: Requirement Analysis

Objective:

Define the technical and functional requirements for the AI board game system.

Key Points:

1. Technical Requirements:

- Programming Language: Python
- AI Framework: TensorFlow/PyTorch (for machine learning)
- Game Integration: Board game APIs or custom game logic
- Frontend: Unity or Pygame (for visual representation)
- Backend: Flask/Django (for AI logic and game state management)

2. Functional Requirements:

- AI opponent that can play multiple board games (e.g., Chess, Checkers, Catan).
- AI advisor that provides real-time strategic advice.
- Adaptive difficulty levels based on player skill.
- Playful AI personality with customizable traits (e.g., humorous, competitive, friendly).

3. Constraints & Challenges:

- Ensuring the AI can handle multiple board game rules and strategies.
- Balancing AI difficulty to provide a fair yet challenging experience.
- Creating a playful and engaging AI personality without being intrusive.

Phase-3: Project Design

Objective:

- Develop the architecture and user flow of the AI board game system.

Key Points:

1. System Architecture Diagram:

- Player interacts with the game via a UI.
- AI system processes the game state and makes decisions (as opponent or advisor).
- AI provides real-time feedback and playful interactions.

2. User Flow:

- Step 1: Player selects a board game and difficulty level.
- Step 2: AI opponent or advisor is initialized based on player choice.
- Step 3: Player interacts with the AI during the game, receiving advice or facing challenges.
- Step 4: AI adapts its strategy and personality based on player performance.

3. UI/UX Considerations:

- Simple, intuitive interface for selecting games and difficulty levels.
- Visual representation of the AI's playful personality (e.g., animations, text bubbles).
- Real-time feedback on AI decisions and advice.

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down development tasks for efficient completion.

SPRINT	TASK	PRIORITY	DURATION	DEADLINE	ASSIGNED TO	DEPENDENCIES	EXPECTED OUTCOME
Sprint 1	Environment Setup&AI	High	6 hours(da	End of Day 1	Member 1	Python,tensor Flow/Pytorch	AI Framework

	Framework Integration		y 1)			Setup	Integrated
Sprint 1	Basic Game Logic Implementation	Medium	2 hours(day 1)	End of Day 1	Member 2	Game Rules Finalized	Basic Game Logic Implemented
Sprint 2	AI Opponent Development	High	3 hours(day 2)	Mid Day 2	Member 1&2	Game Logic, AI Framework Ready	AI Opponent With Basic Decision Making
Sprint 2	AI advisor Development	High	1.5 hours(day 2)	Mid Day 2	Member 1&4	Game logic, AI Framework ready	AI Advisor With Basic Advise Functionality
Sprint 3	Testing and UI Enhancements	Medium	1.5 hours(day 2)	Mid Day 2	Member 2&3	AI logic, UI layout completed	Responsive UI/Better User Exp
Sprint 3	Final Presentation and Deployment	Low	1 hour (day 2)	End Of Day 2	Entire Team	Working prototype	Demo-ready Project

Phase-5: Project Development

Objective:

- Implement core features of the AI board game system.

Key Points:

1. Technology Stack Used:

- Frontend: Unity/Pygame
- Backend: Flask/Django
- AI Framework: Tensor Flow/PyTorch
- Programming Language: Python

2. Development Process:

- Implement AI decision-making logic for board games.
- Develop adaptive difficulty levels based on player performance.
- Create a playful AI personality with customizable traits.

3. Challenges & Fixes:

- Challenge: AI struggles with complex board game strategies.
- Fix: Use reinforcement learning to improve AI decision-making.
- Challenge: Balancing AI difficulty for different skill levels.
- Fix: Implement dynamic difficulty adjustment based on player performance.

Phase-6: Functional & Performance Testing

Objective:

- Ensure that the AI board game system works as expected.

Test case id	category	Test scenario	Expected outcome	Status	Tester
Tc-001	Functional testing	AI opponent plays a game of chess	Ai makes a valid moves and adapts to players skill	Passed	Tester 1
Tc-002	Functional testing	AI advisor provides advice in Catin	Ai offers relevant strategic advice	Passed	Tester 2
Tc-003	Performance testing	Ai response time under 500 ms	Ai response quickly to players move	Needs optimization	Tester 3
Tc-004	Bug fixes&improvents	Fixed incorrect Ai decisions	Ai makes more accurate decisios	Fixed	Developer
Tc-005	Final validation	Ensure Ui responisive across devices	Ui works on mobile and decisions	Failed -UI broken on mobile	Tester 2
Tc006	Deployment testing	Host the Ai systems on a cloud platform	System is accessible online	deployed	devOps

Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link: <https://github.com/PAWAN-5/game-analyzer.git>**
 4. **Presentation**
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